

Exercise 2 – Modular Arithmetic and RSA

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Problem 1. Solve the following equations:

- a) $5x \equiv 6 \pmod{8}$
- b) $7x \equiv 5 \pmod{12}$
- c) $12x \equiv 8 \pmod{20}$

Problem 2. Determine $\varphi(n)$ for the following numbers:

- a) 12
- b) 23
- c) $\triangleleft p^2q$, where p and q are distinct primes.

Problem 3. Compute the following:

- a) $3^{40} \pmod{7}$,
- b) $3^{5^7} \pmod{28}$,
- c) $\triangleleft$ What is the last digit of 1357^{246} ?
- d) $12^{100} \pmod{30}$.

Problem 4. Find all $x \in \mathbb{Z}$ satisfying the following equations:

- a) $4^x \equiv 1 \pmod{7}$,
- b) $5^x \equiv 3 \pmod{13}$,
- c) $2^x \equiv 7 \pmod{15}$.

Problem 5. Prove or disprove the following statement: Let n and b be positive integers. If for every a such that $\text{GCD}(a, n) = 1$ we have $a^b \equiv 1 \pmod{n}$, then $b \equiv 0 \pmod{\varphi(n)}$.

Problem 6. You use the RSA cryptosystem and choose the prime numbers 7 and 11. You choose 13 as the public exponent.

- a) Determine the private and public key.
- b) You receive the encrypted message $c = 14$. Decipher it.

Problem 7. For odd n , prove that $n \mid 2^{n!} - 1$.

Problem 8. Show that every quadratic equation over $\mathbb{Z}_p$, where p is prime, has at most two roots. Explain why this does not hold for arbitrary moduli.

Problem 9. Alice wants to send the same message to Eve and Václav using RSA. Eve's public key is $(N_E = 1073, e = 17)$ and Václav's public key is $(N_V = 1147, e = 17)$. The encrypted messages are 809 and 994, respectively. What message did Alice send?

Homework. The two-digit integers from 19 to 92 are written consecutively to form the large integer N :

$$N = 192021 \dots 909192.$$

Suppose that 3^k is the highest power of 3 that is a factor of N . What is a value of k ?